Run report: Cheon ecbench medium-tier sessions, 2026-11-06T03:00Z

Descriptive only; the interpretation is in ../../REPORT.md.

What ran

Three specs, run in turn with `ecbench run --spec <spec> --out sessions/<name>-20261106 --allow-busy --quiet` at isolation level L0, recording operation counts only. The host is a shared cloud VM with 4 vCPUs, and no wall time is admitted.

session	spec id	session id	arms	curves ($\log_2 r$)	started	finished
minus	ECS1h363c3f2358e9	ECBS1he3401ce241ca	`rho.negation`, `cheon.pminus1_comb`	35.6 to 55.6 (6)	03:00:44Z	03:13:41Z
plus	ECS1h925c4cd120b9	ECBS1h64874297871c	`rho.negation`, `cheon.pplus1_comb`	35.6 to 55.6 (6)	03:13:41Z	03:31:31Z
minus-naive	ECS1h2934f0735cb6	ECBS1hb3aa04e7a9c6	`rho.negation`, `cheon.pminus1_naive`	35.6 to 51.6 (5)	03:31:31Z	03:53:01Z

- **Specs.** `../../spec-*.json`. Each is `research/cheon_ecbench_20261022/spec-*.json` without its 32-bit curve; the arms, shapes, targets (8 per curve, seed 20261022) and measurement block are unchanged.
- **Binary.** sha256 `f561cc6c…`. It was built from the sources of `ac79a4b04f`, which equal those of `b83ccff8ce`, the clean commit checked out at run time. The recorded `git_commit` is stale; see ../../sessions/BINARY.txt.

Completion

session	runs requested	completed	verified	failures / timeouts
minus	96	96	96	none
plus	96	96	96	none
minus-naive	80	80	80	none

Audit

Each session was replayed in full with `ecbench verify --dir <session> --replay-all`.

receipt	runs reproduced	problems	sha256
`sessions/minus-20261106.audit.json`	96 / 96	0	`91b8644598434800d4c5205a6f1df052a58262608ee8afde384c3fefc894a287`
`sessions/plus-20261106.audit.json`	96 / 96	0	`59baa234ed6a2f9d012aaf21f40e1f6c630c36e4f4d8508fb56352277e4862b0`
`sessions/minus-naive-20261106.audit.json`	80 / 80	0	`d76b80e728e415725698b32c80b4dee1100727eba080a62d922483a2eb6027e4`

Against the original sessions

../../compare.sh compares every run with the original session's run of the same arm, curve and target index. The output is ../../results-p1.jsonl. The fields compared are:

- outcome, recovered scalar and `matches_planted`;
- `total_gae` and `S`;

- oracle queries and oracle server work;
- per-phase adds, doubles, scalar multiplications and GAE.

session	runs	identical	absent from the original	differing
minus	96	96	0	0
minus-naive	80	80	0	0
plus	96	96	0	0

Measured values

arm (session)	runs	S (mean, from the bound)	95 %
`cheon.pminus1_comb` (minus)	48	0.574	[0.449, 0.722]
`rho.negation` (minus)	48	0.987	[0.744, 1.231]
`cheon.pminus1_naive` (minus-naive)	40	12.243	[9.382, 14.909]
`rho.negation` (minus-naive)	40	1.055	[0.802, 1.336]
`cheon.pplus1_comb` (plus)	48	0.582	[0.417, 0.772]
`rho.negation` (plus)	48	1.013	[0.758, 1.312]

`S` is the w = 0 constant of each whole-session fit-version-2 bound, from `docs/bounds/records/cheon-ecbench-medium-*.json` (fitted by `../../fit.sh`). The graph is `../../ranks.svg`. Raw records are in `../../sessions/<session>/records.jsonl`.